

M33. INTERFERENCES IN THE QUINCKE TUBE

OBJECTIVES

- Check for the occurrence of constructive and destructive interferences.
- Determination of the sound speed and the characteristic length of the tube.

THEORETICAL BASIS

A very important characteristic of wave motion is the phenomenon of interference. This occurs when two or more wave motions coincide in space and time. In our case we will achieve this by using the Quincke tube (Figure 1): from (1) pressure waves are emitted which separate through an inverse "T" piece (splitter), travel paths of lengths a and b , and interfere giving rise to a new wave. The sound receiver (2) measures this wave.

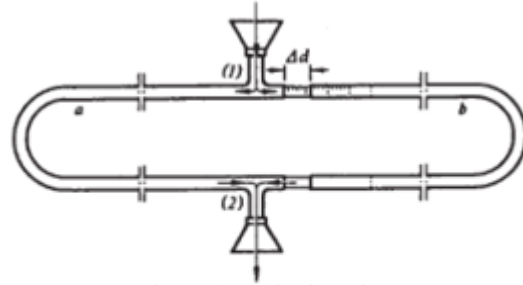


Figure 1. Quincke tube

From Figure 1 it can be deduced that the individual waves do not travel the same distance, their difference being:

$$b - a = 2\Delta d \quad (1)$$

Since both waves come from the same source (1), $s=0$, they will have the same value of intensity, frequency and wavelength, but not the same for their amplitude (due to the intensity-frequency-amplitude relationship, absorption, natural attenuation of spherical waves...). We can represent these waves at their arrival at the receiver (2), $s=s_R$, as:

$$\begin{aligned} p_a(s_R, t) &= p_{0a} \cos(\omega t - ka) \\ p_b(s_R, t) &= p_{0b} \cos(\omega t + kb) \end{aligned} \quad (2)$$

assuming that our emitter emits pure harmonic plane waves that propagate in opposite directions inside the tube, traveling different distances to the receiver. From (2) we have that there exists a phase difference between the two wave motions of value:

$$\delta = k(b + a) = \frac{2\pi}{\lambda}(b + a) \quad (3)$$

Taking into account the principle of superposition, (1) and (3), the amplitude detected by the sound receiver will have the form:

$$p_0 = \sqrt{(p_{0a}^2 + p_{0b}^2 + 2p_{0a}p_{0b} \cos(4\pi(a + \Delta d)/\lambda))} \quad (4)$$

From (4), we know that the total amplitude is between the values of:

$$\begin{cases} \text{maximum amplitude} = p_{0a} + p_{0b} \\ \text{minimum amplitude} = |p_{0a} - p_{0b}| \end{cases} \quad (4.1)$$

depending on:

$$\cos(4\pi f(a + \Delta d)/c) = \begin{cases} +1 \leftrightarrow \Delta d = m \frac{c}{2f} - a \rightarrow p_0 = p_{0a} + p_{0b} \\ -1 \leftrightarrow \Delta d = (2m + 1) \frac{c}{4f} - a \rightarrow p_0 = |p_{0a} - p_{0b}| \end{cases} \quad (5)$$

where it has been taken into account that $\lambda = c/f$ and where m is a natural number. In the case of *maximum amplitude* we have maximum reinforcement of the two wave motions, or *constructive interference*, and in the case of *minimum amplitude* and maximum mitigation, or *destructive interference*. If the distance a is unknown, the two displacement expressions in (5) can be rewritten so that:

$$\Delta d = n \frac{c}{4f} \quad (6)$$

where n is the relative ordinal (for example, $n=19$, first minimum/maximum, $n=20$, adjacent maximum/minimum, $n=21$, second minimum/maximum, $n=22$, adjacent maximum/minimum...).

In the particular case $a=b$ and only for certain frequencies, the interference gives rise to a *standing wave* (resonance) with one antinode in the receiver and another in the transmitter (the Quincke tube is closed on itself). The frequency law that must be fulfilled is:

$$f_n = n \frac{c}{2a} \quad (7)$$

where $n=2, 4, 6 \dots$. The fundamental mode has 4 nodes and 4 antinodes..

METHODOLOGY

Sound waves are generated at position (1) in Figure 1, with the help of a speaker connected to the **Out** output, lower left corner, of the digital function generator. There is an oscilloscope connected to a voltmeter and the function generator which allows the two signals (input and output) to be displayed to identify the maximum and minimum amplitude and phase shift situations. The amplitude of the sound is selected with a dial button (upper right) and the frequency (display) with the main dial button (left), in addition to the range push buttons. The wave type must be sinusoidal. Check with the oscilloscope that the zero offset is at 0.000 V and that the signal amplitude is less than 0.500 V, as the sound generated can be annoying depending on the frequency used. This amplitude should be the same during the same experience (A or B according to the Procedure section). Explore the limiting cases. To measure the amplitude of the sound waves, use a microphone with amplifier placed in position (2) in Figure 1, connected through the oscilloscope to the voltmeter (full scale 2 V AC). The received signal does not need to be amplified (minimum position). In case of amplification, this amplification must be the same during the same experiment (A or B according to the Procedure section).

CAUTIONS

Do not move the speaker and/or microphone during the course of the experiment.

*Check that the level of the sound signal is adequate and does not disturb the rest of the class.
Use your hearing protection.*

Do not vary the amplification during the experiment. Disconnect the microphone source once the experiment is finished.

PROCEDURE

Measure the laboratory temperature.

Experiment A: Determine the frequencies that produce standing waves in the Quincke tube ($a=b$).

- A1. Select a frequency of 1300 Hz and vary the units digit.
- A2. Increase the frequency of the function generator slowly and with the voltmeter (and the help of the oscilloscope) locate an audible maximum. Write down the voltage of this maximum. You may save the oscilloscope image on a USB flash drive connected to the USB input of the oscilloscope.
(<https://www.science.smith.edu/~rolivo/bio330/LAdso-screenshot.html>,
<https://www.youtube.com/watch?v=B0ajIBb4EU4>)
- A3. Repeat A2 until at least a total of 6 consecutive maxima (and 6 minima) are obtained. Capture the most representative screen that appears on the oscilloscope (follow the instructions in the link). List the maxima found as the frequency values increase.

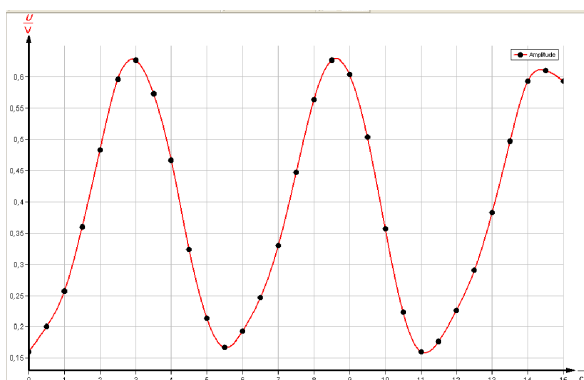
Experiment B: Determine, at fixed frequency, the amplitude of the pressure wave with the displacement of the moving arm of the Quincke tube.

- B1. Select in the function generator the frequency around 1518-1670 Hz for which you observed the maximum in **Experiment A**. Write it down together with the value of the sound amplitude.
- B2. Move the movable arm to 0.5 cm from the top and note the value of the sound amplitude (voltmeter reading).
- B3. Repeat B2 increasing the spacing between the arms of the Quincke tube in 0.5 cm increments until the end of the scale is reached. You should observe an interference pattern with at least 2-3 maxima and 3-2 minima.

Measure the laboratory temperature.

RESULTS

- R1.- Present all the direct results of your experiments (**A** and **B**) in tabular form (units, errors). Accompany them with some oscilloscope screens.
- R2.- Plot the resonance frequency (y-axis) as a function of the order number n (x-axis)



of the maximum (**Experiment A**). Find the best-fit curve from (7). Obtain the c/a value of the slope with its error.

R3.- Plot the sound amplitude as a function of the displacement Δd of the arms of the Quincke tube (**Experiment B**).

R4.- Extract the values of Δd for each maximum and minimum obtained in R3. Plot Δd (y-axis) as a function of the order number n (x-axis) of the corresponding maximum/minimum. Find the best-fit curve from (6). Obtain the value of c of the slope with its error. Compare (relative error) this value of c to the speed of sound at a pressure of 1013 hPa, calculated as $c = 331.3\sqrt{1 + \theta/273.15}$ expressed in m/s and where θ is the temperature in °C.

R5.- Knowing the experimental value of c from R4, express the experimental value of a from the result of R2.

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Modification date

2024-03-19

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